3 **(a)** $\lambda' = 191.9 \times 10^{-12} + \frac{6.63\times10^{-34}}{(9.11\times10^{-31})(3.00\times10^{8})}(1 - cos\ cos\ 64.3^{\circ})$ M1

$\lambda' = 193.3 \times 10^{-12}$ m A0

(b) decrease in energy of photon ΔE
$= (6.63 \times 10^{-34})(3.00 \times 10^{8})(1 / 191.9 \times 10^{-12} - 1 / 193.3 \times 10^{-12})$ C1
$= 7.5068 \times 10^{-18}$ J

decrease in energy of photon transfer to kinetic store of electron or $\Delta E = P^2 / 2m$ B1

$P = [2\ (9.11 \times 10^{-31})\ (7.5068 \times 10^{-18})]^{1/2}$ M1

$= 3.698 \times 10^{-24}$ N s A0

(c) in direction perpendicular to incident photon,
momentum of deflected photon = $(6.63 \times 10^{-34} / 193.3 \times 10^{-12})$ sin 64.3° C1

momentum of deflected electron = 3.698×10^{-24} sin α C1

clear statement stating conservation of momentum at right angle to the incident photon (e.g. incident photon has no momentum perpendicular to direction of incidence, so perpendicular component of electron's momentum equal and opposite to that of the deflected photon) M1

3.698×10^{-24} sin $\alpha = (6.63 \times 10^{-34} / 193.3 \times 10^{-12})$ sin 64.3°

α = 56.7° or 56.69° A1